1

Version 1.0 | 2026-08-11

Repository: github.com/sykeisuke/asic_rd

ASIC SPICE RTL GDSII GitHub

4

- GF180 PDK MOS
- Wilkinson ADC
- SPICE RTL Gray code CDC
- RTL-to-GDS DRC/LVS/STA
- prototype tape-out-ready design

0.

ASIC software process 3 log

4

/ schematic

netlist

RTL

layout

model

- SPICE model transistor nominal simulation silicon
- RTL simulation logic clock transistor-level analog
- GDS package bond wire PCB system

m
μ / u
n
p

Lab commit hash command PASS/FAIL artifact

1.

3 tool Makefile command

Tool
Docker Desktop
Make
Xschem
ngspice
AWK
Icarus Verilog
Yosys
OpenSTA
LibreLane
KLayout/Magic/Netgen

KLayout/Magic/Netgen analog layout digital physical flow LibreLane bac

2. Makefile

make simulator Makefile target shell script make nmos-dc scripts/run-nmos-dc.sh script Docker Xschem ngspice

make nmos-dc -> scripts/run-nmos-dc.sh -> docker run -v <repo>:/foss/designs -> xschem -n ... nmos_dc.sch -> ngspice -b ...spice -> work/nmos_id_vds.csv

script

1. set -eu error
2. eda-common.sh EDA_IMAGE PROJECT_ROOT Docker CLI
3. docker run -v Mac repo container /foss/designs
4. container cd
5. tool option -b batch -o log
6. test/grep PASS

Target
make nmos-dc
make four-cell-cosim
make digital-top
make digital-physical

make target scripts/run-*.sh command command

3.

4 device bias sampling

digital 0/1 analog node DC operating point signal large transient noise/mismatch

3.1 MOS transistor

VGS
VDS
ID
gm

ro

W/L

body MOS 4 device bulk/body body effect parasitic diode

3.2 Bias headroom gain bandwidth

Bias signal transistor node Headroom signal device

- DC operating point: node branch
- Gain:
- Bandwidth: gain
- Slew rate: large signal
- Settling:

RC sampling

switch ON RON hold capacitance C τ = RON·C exp(−t/τ) sampling pulse hold node settle C kT/C noise droop acquisition

C

RON = 2 kΩ C = 1 pF τ = 2 ns 0.1% settle 7τ 14 ns first-order signal-dependent RON SPICE

3.3 noise PVT Monte Carlo

Systematic error

Random noise

Mismatch

PVT

Parasitic

Nominal simulation typical model Tape-out fast/slow device corner min/max mismatch seed extracted parasitic verification plan

1. ground
2.
3. node rail
4.
5.
6. cursor transient

4.

5 module controller analog crossing Gray capture code

Signal
VIN
SAMPLE[3:0]
VHOLD[3:0]
SEL[1:0]
VMUX
VRAMP
HIT
CODE[5:0]
DATA

4.1 1

1. Track: SAMPLE[n]=1 switch ON VHOLD[n] VIN
2. Hold: SAMPLE[n]=0 switch OFF capacitor
3. Select: SEL=n VHOLD[n] VMUX
4. Settle: MUX transient
5. Reset: VRAMP counter
6. Convert: ramp Gray counter
7. Compare: VRAMP VMUX HIT edge
8. Capture: HIT Gray code binary code

9. Repeat: n=0..3

10. Readout: 4×6 bit 24-bit word serial

analog/digital

Comparator → capture
Controller → analog switch
Code RAM → serializer

arrow producer consumer voltage domain clock domain active polarity reset state

5.

2 GDSII complete flow

Sampling cell
Analog MUX
Ramp
Comparator
Gray counter/capture
Serializer

4 storage cells 6-bit ramp/comparator 24-bit serial readout flow

Lab 0

git clone https://github.com/sykeisuke/asic_rd.git cd asic_rd git status make check

5

README.md
docs/PROTOTYPE_SPECIFICATION.md
docs/TOP_LEVEL_INTERFACE.md
docs/VERIFICATION_MATRIX.md

docs/TAPEOUT_BLOCKERS.md

- simulations/: Xschem/ngspice testbench
- digital/: synthesizable RTL testbench physical design
- docs/:
- work/ runs/: source of truth

- HEAD commit
- VERIFICATION_MATRIX 1 test artifact

6. Mac ✦ Docker

Mac Linux EDA GF180 PDK Docker Xschem GUI VNC macOS

1. Docker Desktop Engine Running
2. make vnc
3. http://localhost:8080/?password=abc123
4. File Manager /foss/designs/simulations/gf180_nmos_dc/nmos_dc.sch

cd /Users/ykeisuke/Desktop/mywork/asic_rd make vnc # Browser: http://localhost:8080/?password=abc123

image Docker Hub private image rate limit image digest tag

---

localhost:8080

Xschem PDK symbol

com.docker.vmnetd

Gatekeeper Docker helper cleanup

6.1 Docker VNC /foss

VNC Docker Desktop Docker Desktop IIC-OSIC-TOOLS Linux container desktop Browser localhost:8080 container web/VNC server Linux desktop

Mac -> Docker Desktop -> container: asic-rd-gf180-vnc -> Linux desktop + EDA tools + PDK -> web/VNC server : port 80 -> Browser localhost:8080

IIC-OSIC-TOOLS

application

/foss

Connected to <id>

volume mount

eda-vnc.sh -v "$PROJECT_ROOT:/foss/designs:rw" PROJECT_ROOT make vnc asic_rd repository

Student Mac: /Users/.../asic_rd | Docker bind mount Container: /foss/designs

git clone asic_rd directory cd make vnc /foss/designs clone mount

7. Xschem

Xschem editor simulation engine symbol wire SPICE netlist netlist ngspice project GUI script headless netlisting

1. make vnc browser desktop
2. File Manager /foss/designs/simulations/gf180_nmos_dc/
3. nmos_dc.sch double-click Xschem File > Open
4. NMOS symbol voltage source ground simulation command
5. symbol property model W L multiplicity

- junction dot

- node label net
- ground/reference node SPICE
- MOS D/G/S/B symbol property/netlist

- testbench copy Git branch
- W L bias stimulus
-
- git diff

Xschem shortcut version keymap menu status/help s

7.1 nmos_dc.sch

6 nmos_dc.sch graph RAW

7.2

1. Mac terminal repository make nmos-dc work/nmos_dc.raw
2. Xschem Load simulation results Ctrl +
   launcher property

GUI Netlist Simulate launcher Ctrl +
Xschem $netlist_dir ngspice write work/ write nmos_dc.raw

3. graph Id-Vds curve
4. curve Xschem directory /foss/designs/simulations/gf180_nmos_dc
5. Terminal ls -lh work/nmos_dc.raw file 0 byte
6. make nmos-dc Xschem Ctrl +

Mac terminal cd /Users/ykeisuke/Desktop/mywork/asic_rd make nmos-dc # Container terminal cd /foss/designs/simulations/gf180_nmos_dc ls -lh work/nmos_dc.raw

R W raw_read(): failed to open ... Xschem /foss/designs/simulations/gf180_nmos_dc

7 Id–Vds VGS curve 520 µA

- VGS curve drain current
- VDS VDS VDS
- u µA 800u 800 µA
- SPICE source NMOS drain current i(vd) −1 * Xschem graph RPN
- graph curve VGS sweep
- typical model nominal DC sweep PVT

RAW curve VGS/VDS/ID/W/L/model corner Lab 1

7.3 Netlist ngspice

make nmos-dc Xschem nmos_dc.sch work/nmos_dc_xschem.spice ngspice batch mode netlist raw data CSV

xschem -n -q -x --rcfile <gf180-xschemrc> \

- -o work -N nmos_dc_xschem.spice nmos_dc.sch ngspice -b work/nmos_dc_xschem.spice

Option
-n
-q
-x
--rcfile
-o work
-N file
ngspice -b

1. terminal code 0
2. work/nmos_dc_xschem.spice device instance model include
3. work/nmos_dc.raw

4. work/nmos_id_vds.csv header
5. directory README.md RESULTS.md expected result

generated netlist netlist model node parameter

8. Digital tool

Compile
Run
Waveform
Synthesis
Timing
Physical

VCD

1. clock reset
2. controller state cell select hit captured code
3. cursor capture edge edge
4. unknown X impedance Z glitch
5. serializer clock/data/valid timebase

Synthesis

testbench #delay display simulation real model silicon Yosys source list run-digital-top.sh mapped netlist GF180 standard-cell instance

Lab 1 GF180 MOS

NMOS I–V bias

make nmos-dc

1. nmos_dc.sch device W/L body supply
2. testbench VGS/VDS sweep
3. make nmos-dc CSV/plot
4. VGS VDS linear saturation

□ ngspice error

□ simulations/gf180_nmos_dc/work/nmos_id_vds.csv

□ sweep

process corner device option MPW provider

W 2 variant bias Id simulation

Lab 2 Sampling cell

track/hold acquisition droop charge injection

make sampling-cell make four-cell make four-cell-mux

sampling_cell.sch Xschem GUI Netlist Simulate Load sampling waveforms Ctrl + VIN VHOLD SAMPLE 3 SAMPLE high VHOLD VIN low

.sch graph launcher GUI RAW path Make target Xschem GUI

1. SAMPLE high VHOLD VIN
2. SAMPLE edge step charge injection/feedthrough
3. hold droop rate [V/s]
4. 4-cell test cell
5. MUX cell

Cell
0
1
2
3

prototype testbench nominal PVT/Monte Carlo

- cell sampling pulse node naming
- MUX bus SEL decode transmission gate complementary control
- capacitance leakage path simulation timestep

Lab 3 Comparator ramp

ADC

make comparator make comparator-range make comparator-offset make ramp-generator

Wilkinson ramp counter capture t_cross = (V_hold − V_start) / slope ramp slope offset delay noise code

Test
comparator
range
offset
ramp

6-bit

VFS LSB VFS/64 1.6 V 25 mV/LSB prototype code ordering monotonicity INL/DNL

ramp slope ±10% crossing time code

Lab 4 1-cell Wilkinson slice

sampling ramp comparator counter 1

make wilkinson-slice make transfer

1. sampling phase input hold capacitor
2. conversion start ramp reset counter
3. VRAMP VHOLD comparator edge
4. edge counter capture end-of-conversion
5. input sweep code transfer

□ code

□ input code

□ 0–63

□ dead region

transient capture 1 timestep max timestep code

Lab 5 4-cell shared conversion

4 1 ramp/comparator 6-bit code

make four-cell-wilkinson

G W

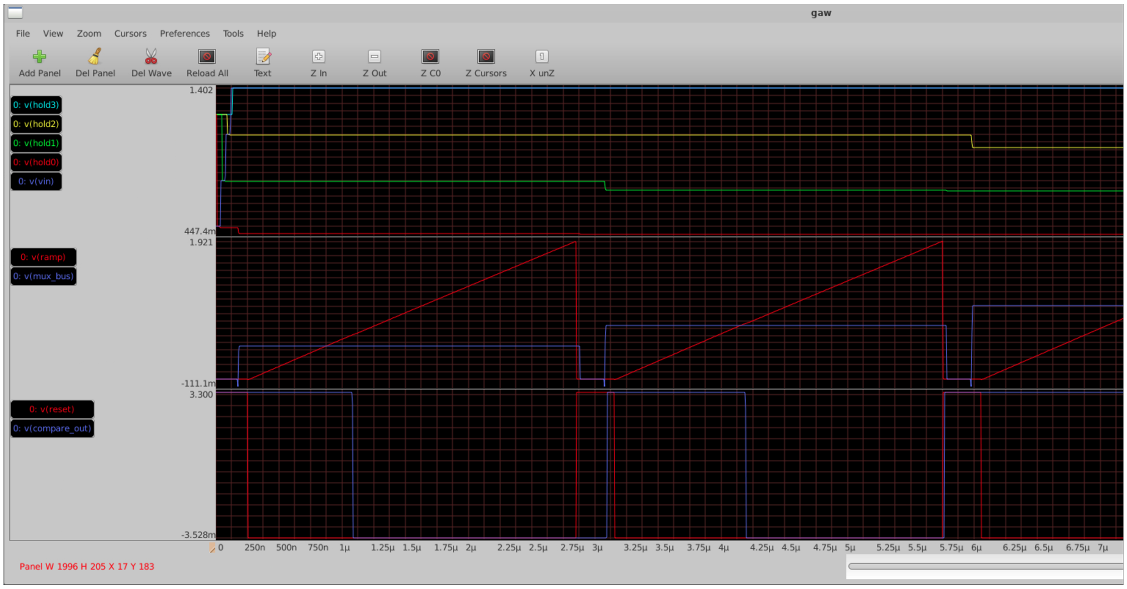
make four-cell-wilkinson GAW RAW VNC Linux Terminal

```
cd /foss/designs/simulations/gf180_four_cell_wilkinson
gaw work/four_cell_wilkinson.raw
```

GAW graph panel drag and drop 3

- : v(vin) v(hold0) v(hold1) v(hold2) v(hold3)
- : v(mux_bus) v(ramp)
- : v(reset) v(compare_out)

window Z Out Zoom menu
4 cell simulation 11.6 µs



4 GAW 4-cell SPICE 4 MUX bus ramp reset comparator 7 μs cell Zoom Out

1. hold0 hold3
2. mux_bus slot hold
3. reset ramp slot
4. ramp mux_bus compare_out
5. hold 6-bit code

Terminal PASS: four cells complete sequential Wilkinson conversion work/measurements.txt code0 code3

3 4-cell shared Wilkinson conversion MUX bus ramp comparator

Cell
0
1
2
3

code 16 < 20 < 27 < 35 sample ⟶ select ⟶ compare ⟶ capture

MUX settling cell crossing

Lab 6 Controller RTL

4-cell sequencing RTL

make counter make gray-counter make controller make digital-top

1. IDLE: start valid flag clear
2. RESET_RAMP: ramp counter
3. SELECT: cell MUX settling
4. CONVERT: Gray counter comparator hit
5. CAPTURE: code register
6. NEXT/DONE: 4 cell serializer

RTL test

- reset X
- cell select one-hot binary
- capture cell
- timeout state machine
- done data_valid pulse/level testbench

simulation delay initial real synthesizable core analog testbench

Lab 7 SPICE-to-RTL

testbench end-to-end code

make four-cell-cosim

SPICE crossing event RTL testbench simulator RTL simulator AMS

contract

---

time unit
edge polarity
sampling convention
code format
invalid case

debug

1. SPICE CSV

2. event
3. testbench pulse
4. capture register enable
5. expected code ±1 sequence

Lab 8 Gray capture CDC

comparator edge counter clock

make phase-sweep

4 cell 2 phase sweep +500 ps code 27 28

binary counter 011 111⟶100 000 bit edge capture code Gray counter code 1 bit code

- +400 ps 27 +600 ps 28 ±1 code
- +500 ps cell 2 boundary setup/hold
- silicon metastability clock skew comparator delay

docs/decisions/0003-gray-comparator-capture.md

9. 24-bit serial readout

4 6-bit code 24-bit word FPGA serial pad bit order frame clock domain idle level

make digital-top

Field
Payload
Bit order
Handshake
Clock
Reset

logic analyzer

1. code 16, 20, 27, 35 load
2. serial clock data

3. 24 edge decode 4 code

4. frame bit slip

PCB I/O voltage drive strength level shifting ground reference connector pinout pad library

Lab 9 RTL-to-GDS

floorplan placement CTS routing signoff

make digital-physical

5 GF180 standard-cell flow top

Metric
Core/die size
Die area
Utilization
Standard cells
Total wire / vias
Estimated power

digital/asic_digital_top/physical/final_views/ GDS DEF LEF gate-level netlist metrics

10. Signoff

Check
DRC
Antenna
LVS
Setup
Hold
Worst setup slack
Worst hold slack

0 violations tape-out ready library/view constraint

- metrics.json/csv:
- asic_digital_top.gds: foundry

- asic_digital_top.nl.v: gate-level netlist
- resolved.json: flow
- run log/report: warning waiver

- clock period setup slack
- hold slack corner OCV I/O constraint

11.

debug loop

1. commit command
2. testbench cell
3. input internal state output timebase
4. 1 1
5. 1
6. test test
7. commit decision log

SPICE convergence
code 0/63
cell
±1
RTL sim PASS P&R
DRC/LVS

12.

- make check make four-cell-wilkinson
- commit hash command 2
- nominal

B DC

- ramp slope conversion time code scale
- simulation
- VERIFICATION_MATRIX test 1

C CDC boundary

- phase sweep
- binary capture Gray capture failure mode
- ±1 code

D physical experiment

- clock constraint utilization 1
- area slack wire length runtime
- metric metric

README diff command artifact path

13. Prototype tape-out

complete design flow demonstration foundry full chip GDS provider

MPW/provider
Analog layout
PVT/Monte Carlo
Top integration
Pad/ESD
Package/PCB
Full-chip signoff

flow GF180 gf180mcu_fd_sc_mcu7t5v0 timing views pad library M

Tape-out readiness review

□ PDK tool version

□ test artifact

- ☐ waiver owner
- ☐ pad ring seal ring density/fill top GDS
- ☐ package/PCB/test plan chip pinout

14. Specification snapshot

Process
Channels
Storage depth
ADC
Readout
Control
Primary objective
Not yet claimed

PDK
MPW
SCA
Wilkinson ADC
CDC
DRC
LVS
STA
PEX
GDSII

15. Command quick reference

Command
make check
make nmos-dc
make sampling-cell
make four-cell
make comparator

make comparator-range
make comparator-offset
make ramp-generator
make wilkinson-slice
make transfer
make four-cell-mux
make four-cell-wilkinson
make gray-counter
make controller
make four-cell-cosim
make phase-sweep
make digital-top
make digital-physical
make vnc

16. Final review sheet

README prototype complete flow

□ 4-cell/6-bit

□ sampling error ADC code

□ ramp/comparator/counter

□ Gray capture

□ SPICE RTL contract

□ DRC/LVS/STA 0 violations

□ source generated artifact

□ tape-out top-level blocker

analog layout + extracted simulation sampling cell layout schematic → layout

- GitHub: https://github.com/sykeisuke/asic_rd
- docs/PROTOTYPE_SPECIFICATION.md
- docs/VERIFICATION_MATRIX.md
- docs/TAPEOUT_BLOCKERS.md

- docs/decisions/0003-gray-comparator-capture.md

command commit